

torch.autograd.graph.Node.register_hook

https://pytorch.org/docs/stable/generated/torch.autograd.graph.Node.register_hook.html

Description:

Register a backward hook. The hook will be called every time a gradient with respect to the Node is computed. The hook should have the following signature: The hook should not modify its argument, but it can optionally return a new gradient which will be used in place of grad_inputs. This function returns a handle with a method handle.remove() that removes the hook from the module.

Function Signature

```
abstract Node.register_hook(fn)[source]#
```

Return type

Returns:

RemovableHandle

Code Examples

```
hook(grad_inputs: Tuple[Tensor], grad_outputs: Tuple[Tensor]) ->
Tuple[Tensor] or None
```

```
>>> import torch
>>> a = torch.tensor([0., 0., 0.], requires_grad=True)
>>> b = a.clone()
>>> assert isinstance(b.grad_fn, torch.autograd.graph.Node)
>>> handle = b.grad_fn.register_hook(lambda gI, gO: (gO[0] *
2,))
>>> b.sum().backward(retain_graph=True)
>>> print(a.grad)
tensor([2., 2., 2.])
>>> handle.remove() # Removes the hook
>>> a.grad = None
>>> b.sum().backward(retain_graph=True)
>>> print(a.grad)
tensor([1., 1., 1.])
```

Notes & Warnings

NOTE

Note See Backward Hooks execution for more information on how when this hook is executed, and how its execution is ordered relative to other hooks.

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Detailed Documentation

Documentation

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The hook should not modify its argument, but it can optionally return a new gradient which will be used in place of `grad_inputs`.

This function returns a handle with a method `handle.remove()` that removes the hook from the module.

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